(iii) Amount, $A = P\left(1+\frac{a}{100}\right)\left(1+\frac{b}{100}\right)$

$= 30000\left(1+\frac{7}{100}\right)\left(1+\frac{8}{100}\right)$

$= 30000\times\frac{107}{100}\times\frac{108}{100}$

$A = ₹34668$

$\therefore$ C.I = A − P = 34668 − 30000 = ₹4668

(iv) Amount, $A = P\left(1+\frac{r}{100}\right)^{a}\left(1+\frac{\frac{b}{c}\times r}{100}\right) = 10000\left(1+\frac{8}{100}\right)^{2}\left(1+\frac{\frac{3}{4}\times 8}{100}\right)$

$= 10000\left(\frac{27}{25}\right)^{2}\left(\frac{53}{50}\right)$

$= 10000\times\frac{27}{25}\times\frac{27}{25}\times\frac{53}{50}$

$A = ₹12363.84$

$\therefore$ C.I = A − P $= 12363.84 - 10000 = ₹2363.84$

4.4.1 Applications of Compound Interest Formula

The compound interest formula is used in the following situations.

(i) To find the increase $\left[P\left(1+\frac{r}{100}\right)^{n}\right]$ or decrease $\left[P\left(1-\frac{r}{100}\right)^{n}\right]$ in population.

(ii) To find the growth of cells when the rate of growth is given.

(iii) To find the depreciation in the values of machines, vehicles, utility appliances etc.,

Example 4.15

The value of a motor cycle 2 years ago was ₹70000. It depreciates at the rate of 4% p.a. Find its present value.

Solution:

Depreciated value $= P\left(1-\frac{r}{100}\right)^{n}$

$= 70000\left(1-\frac{4}{100}\right)^{2}$

$= 70000\times\frac{96}{100}\times\frac{96}{100}$

$= ₹64512$



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